

A High-Reliability Sustained Power Supply System for Polar Regions Based on Hybrid

Energy Storage Technology: Analysis, Design, and Field Trials

Abstract:

Seismic exploration is of significant scientific importance for revealing the geological structure and resource distribution beneath the Antarctic ice sheet. However, existing field instruments face challenges in achieving long-term stable power supply, particularly in the extreme Antarctic environment characterized by severe cold and highly fluctuating renewable energy sources. Conventional energy storage systems struggle with insufficient energy density and performance degradation at low temperatures. To address these issues, this study proposes a hybrid wind-solar power supply system based on a supercapacitor and lithium battery combination, along with a multi-mode cooperative control strategy adapted to low-temperature conditions. The system employs a parallel architecture of photovoltaic panels and vertical-axis wind turbines, integrating constant voltage/constant current control to achieve adaptive energy distribution. Field tests conducted in Antarctica, with over 100 hours of operation, demonstrate that the system achieves a low-temperature startup response time of less than 30 seconds, improves power supply continuity by 7.25 times. The results confirm the system's capability to reliably support the long-term operation of small distributed seismic acquisition stations in polar regions, demonstrating promising prospects for engineering applications. This research offers novel insights and technical references for the design of energy systems in complex polar environments.

I. Introduction

As a crucial component of the Earth's system [1], the Antarctic continent holds irreplaceable strategic value in the study of global climate evolution, resource distribution, and geological activities. However, constrained by extreme climatic conditions and environmental inaccessibility, Antarctica remains one of the least understood regions in Earth science research[2], [3]. Therefore, advancing research on geophysical exploration technologies tailored to polar environments, particularly seismic exploration techniques, is of great significance for enhancing our understanding of key geological structures, including subglacial lakes, geological fault zones, and groundwater systems.

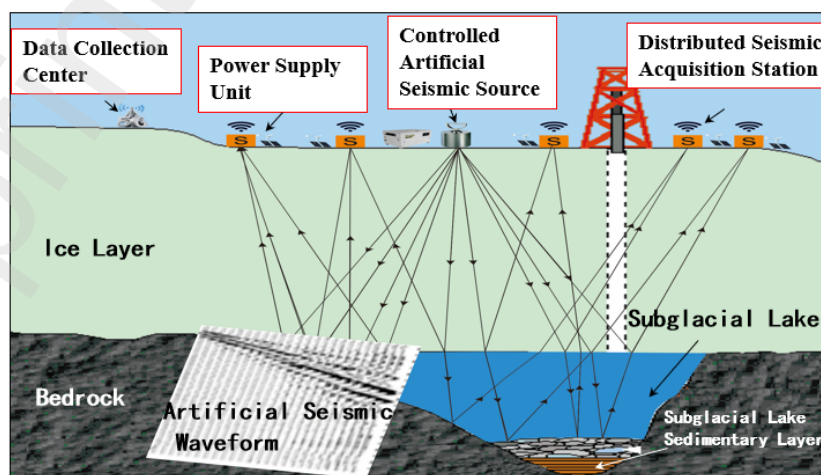


Fig 1. Schematic diagram of subglacial lake detection using an active-source seismic method

Ultra-long-term active-source seismic exploration combined with seismomagnetic monitoring holds profound significance for scientific research and practical applications[3]. It offers high precision and resolution, with current detection depths ranging from tens of meters to several tens of kilometers[4]. A key characteristic is that all generated vibration frequencies can propagate through geological strata with effective energy conservation and minimal disturbance to rocks and the ground[5]. Moreover, this method exhibits strong anti-interference capability and a high signal-to-noise ratio. Artificial seismicity has broad applications in seismic exploration, active fault detection, nuclear explosion monitoring, and volcanic activity detection[6]. By observing and analyzing the subsurface medium's response to artificially induced seismic waves, the properties and geometry of target structures can be inferred.

In Antarctica, active-source seismic exploration enables continuous observation of ice sheet vibrations across different periods. By analyzing the physical mechanisms of these vibrations, the interface between the ice layer and subglacial lakes can be clearly identified, allowing for precise determination of their locations and characteristics[6].

Due to its advantages such as controllable energy, high resolution, and strong anti-interference performance, active-source seismic technology is widely used in polar regions[7]. It is particularly suitable for high-resolution detection of multi-scale structures beneath ice sheets[8]. Through the deployment of distributed seismic acquisition stations in the field, long-term and continuous recording of ice sheet vibrations across different frequency bands can be achieved, facilitating the inversion of basal ice sheet structures and subglacial lake boundary features[3]. Previous studies have demonstrated the method's promising potential in monitoring active faults, early warning of volcanic activity, and studying ice sheet dynamics.

Distributed seismic acquisition stations are key components of active-source seismic exploration equipment, as illustrated in Fig 1. They offer high mobility and flexibility, enabling rolling acquisition of seismic signals, which makes them particularly suitable for exploration in logistically challenging environments such as Antarctica. Current mainstream distributed seismic acquisition systems (e.g., SUMMIT X One, SmartSolo, GSX, etc.) are typically powered by lithium batteries, with power consumption generally ranging from 1.2 W to 6 W, providing a theoretical continuous operation time of 600 to 800 hours. [9].

TABLE I Comparison and Analysis of Power Supply Schemes and Operating Durations

Name	Geode DZ	ES3000	SmartSeisST	StrataVisor NZXP	[9]	[8]
Power Supply Type	Lithium-Ion Battery	Lithium-Ion Battery	Lithium-Ion Battery	Lithium-Ion Battery	Lithium-Ion Battery	Lithium-Ion Battery
Power Consumption	0.5W/channel	0.65W /channel	26W+0.65W per channel	0.5W/channel	1.2W	6.646W/0.0 24W
Expected Operating Duration	720h	600h	720h	800h	800h	600h

However, the power supply for distributed seismic acquisition stations poses a significant challenge for researchers under the unique Antarctic environment. A typical active-source seismic

survey requires the deployment of dozens of acquisition stations, often situated in rugged, inaccessible terrain where generator use is impractical. Constraints such as cabling limitations, frequent plugging and unplugging, and issues like equipment wear, carbon buildup, port mismatches, and electrical sparks further reduce system reliability.

Current long-term observation efforts reveal the following challenges:

1. Extreme low temperatures impair energy storage efficiency: Batteries that operate for up to 800 hours under normal temperatures exhibit significantly shorter lifespans in extreme cold. During the Antarctic winter, temperatures can drop to -80°C , causing substantial capacity degradation in lithium batteries, while traditional lead-acid batteries are prone to freezing, drastically reducing equipment endurance.

2. Difficulty in energy replenishment: Remote stations often rely on diesel generators; however, Antarctic logistics are extremely costly, fuel reserves are limited, and supply chains are frequently disrupted by blizzards.

3. Limited utilization of renewable energy: Although some stations have experimented with solar or wind power, the polar night leads to insufficient sunlight, and strong winds can damage equipment. Renewable sources currently constitute less than 30% of the energy mix, failing to meet annual demand.

To further contextualize the power supply landscape for Antarctic research, Table 2 summarizes the prevailing power supply configurations and their applications across Antarctic stations. Specifically, Japan's Antarctic Syowa Station employs a multi-source setup integrating three 100 kW diesel generators, a 100 kW photovoltaic (PV) system, and five 20 kW wind turbines. For China's Taishan Station, a wind-solar-diesel-battery co-generation system is adopted, consisting of a 102.4 kW PV array, a 250 kW wind power system, and a 150 kW diesel generator. Meanwhile, a 30 W proton exchange membrane fuel cell serves as the power source for China's Qinling Station. Beyond these station-focused solutions, a 200 W wind-PV hybrid system is proposed for Antarctic subglacial lake operations.

As noted, while large-scale power systems have been widely explored, existing configurations remain limited in addressing the specific needs of portable field equipment. Most portable instruments still rely on batteries, and dedicated power solutions for long-duration field operations (e.g., seismic acquisition stations) are underdeveloped. This gap, compounded by the Antarctic environment's high variability (which renders standalone renewable energy insufficient for stable power supply), underscores the urgency of advancing both portable power systems and supporting energy storage technologies tailored to field scenarios.

Table II Summary of Existing Power Supply Modes for Antarctic Research Stations

Name	[7]	[10]	[11]	proposed
Power Supply Type	diesel generators+PV+wind	wind-solar-diesel-battery co-generation	Proton exchange membrane fuel cell	Wind-PV Hybrid System
Power Range	Three 100kW diesel generators + 100kW PV + five 20kW wind turbines	PV (102.4kW) +wind (250kW) +	30W	PV(150W)+wind(50W)

		diesel (150kW)		
Application Area	Japan's Antarctic Syowa Station	China's Taishan Station	China's Qinling Station	Antarctic Subglacial Lakes

Based on the above background, this paper proposes a sustained high-reliability power supply device for polar long-term operations, utilizing hybrid energy storage technology. The system adopts a hybrid energy storage architecture combining supercapacitors and lithium batteries, establishing a multi-source cooperative management strategy and an extreme-cold start mechanism to enhance stability, durability, and practicality under harsh conditions. Compared with traditional single-battery solutions, the system employs a DC/DC parallel control strategy to enable dynamic power superposition from wind and solar sources.. Integrated with seamless switching and preheating control mechanisms, it effectively enhances cold-start capability below -30°C and ensures sustained power supply performance. The related design has been verified through simulation and field tests, demonstrating promising engineering adaptability and potential for broader application.

The main contributions of this paper are as follows:

1. A multi-source parallel input system is established at the power input end, where solar panels and wind turbines achieve complementary power output via a coordinated operation strategy. The ampere-hour integral method is used to estimate the SOC of the lithium battery. Here, a parallel-connected DC/DC converter topology is adopted for power regulation, achieving effective power superposition. It also effectively avoids current-sharing failures caused by voltage loop competition in multiple constant-voltage power supplies operating in parallel.

2.A management system is employed at the battery end to simultaneously monitor the supercapacitor and lithium battery. The ampere-hour integral method estimates the SOC of both, and usage of a battery is suspended when temperatures become excessively high, thereby improving stability and safety.

3.A hybrid output system comprising supercapacitors and lithium batteries is constructed at the output end. To address performance degradation of batteries at low temperatures, supercapacitors are used for preheating and supplying high current. Once a stable temperature is reached, the system switches to battery power. When both sources operate in parallel, only the side with higher voltage maintains output characteristics. A seamless battery switching control method is proposed, effectively preventing voltage drops during battery transitions.

The structure of this paper is organized as follows: Chapter II analyzes the theoretical foundations of the technology. Chapter III introduces the system composition and overall design. Chapter IV presents the key technologies of the high-efficiency power supply system based on supercapacitors and lithium batteries. Chapter V describes the corresponding experiments. The final chapter summarizes the technology.

II. Current Requirements for Long-Term Power Supply Systems

2.1 Analysis of Power Supply System Requirements

The power requirements of the supply system primarily stem from two components: thermal losses for equipment heating and the operational power consumption of small distributed acquisition stations. A detailed assessment is provided below:

1. Thermal power loss is calculated based on the theory of steady-state heat conduction using the flat-plate formula:

$$Q = kA \frac{\Delta T}{d} \#(1)$$

where:

Q represents the thermal power loss (unit: W); k denotes the thermal conductivity coefficient of the insulation material (unit: W/(m·K)); A is the surface area of the insulated enclosure (unit: m²); ΔT is the temperature difference between the interior and exterior environments (unit: K);

D indicates the thickness of the insulation layer (unit: m).

The load of small distributed acquisition stations is mainly attributed to the power consumption of internal data acquisition circuits, with an operational power consumption P typically ranging between 1 W and 6 W.

The total power requirement of the supply system is the sum of the thermal power loss and the operational power consumption of the acquisition station, expressed as:

$$P_{total} = Q + P \#(2)$$

Based on typical parameter values, the total output power is estimated to be approximately 38 W.

The charging system incorporates a complementary solar and wind power solution. Its output power is determined using the fundamental electrical power formula:

$$P_{out} = UI \#(3)$$

where:

P represents the average output power of the hybrid solar-wind charging system (unit: W);

U is the output voltage of the charging system (unit: V);

I denotes the average output current of the charging system (unit: A).

2.2 Calculation of Wind-Solar Hybrid Input Power

The power supply capacity of the wind-solar hybrid charging system is central to ensuring the continuous and stable operation of the load. Its total output power is determined by the combined output of the solar power generation unit and the wind turbine, while also being influenced by environmental factors, equipment characteristics, and energy conversion efficiencies. The detailed assessment is as follows:

2.2.1 Evaluation of Solar Power Generation Unit Output

The theoretical maximum output power of a PV module is determined by solar irradiance, module area, and photoelectric conversion efficiency. The fundamental formula based on the photovoltaic effect is:

$$P_{PV,th} = G \cdot S \cdot \eta_{PV,0} \#(4)$$

Where:

$P_{PV,th}$ is the theoretical maximum output power of the PV module (Unit: W); G is the solar irradiance (Unit: W/m²); S is the effective light-receiving area of the PV module (Unit: m²);

$\eta_{PV,0}$ is the photoelectric conversion efficiency of the PV module under Standard Test Conditions (STC: Irradiance 1000 W/m², Cell Temperature 25°C, AM1.5 spectrum).

The photoelectric conversion efficiency of a PV module decreases as the cell temperature rises, necessitating the introduction of a temperature correction coefficient. The temperature correction formula is:

$$\eta_{PV,T} = \eta_{PV,0} \cdot [1 - \alpha_T \cdot (T_{cell} - T_{STC})] \quad \#(5)$$

Where: $\eta_{PV,T}$ is the photoelectric conversion efficiency at the actual cell temperature; α_T is the temperature coefficient of the PV module (Unit: K⁻¹); T_{cell} is the actual operating temperature of the PV module (Unit: K); T_{STC} is the cell temperature under standard test conditions (298.15 K).

Combining equations (1) and (2), the temperature-corrected output power of the PV module is obtained:

$$P_{PV,T} = G \cdot S \cdot \eta_{PV,0} \cdot [1 - \alpha_T \cdot (T_{cell} - T_{STC})] \quad \#(6)$$

2.2.2 Evaluation of Wind Turbine Output Power

The output power of a wind turbine is closely related to wind speed. Its calculation is based on aerodynamic principles and the wind turbine's power curve, incorporating energy conversion efficiencies in a multi-stage process.

According to aerodynamic theory, the maximum wind power captured by the rotor per unit time is:

$$p_m = \frac{1}{2} C_p \rho \pi R^2 v^3 \quad \#(7)$$

Where:

C_p is the power coefficient; ρ is the air density in the actual working environment (Unit: kg/m³); v is the wind speed in the actual working environment (Unit: m/s); R is the radius of the rotor (Unit: m).

The wind energy captured by the rotor is transmitted through a drive system to the generator, where it is converted into electrical energy. Losses occur during transmission and generation. Therefore, the actual output power of the wind turbine is:

$$P_{WT,act} = \frac{\pi \rho D^2 C_p^{max} v^3}{8} \cdot \eta_{trans} \cdot \eta_{gen} \quad \#(8)$$

Where:

$P_{WT,act}$ is the actual output power of the wind turbine (Unit: W);

D is the rotor diameter ($D = 2R$) (Unit: m); C_p^{max} is the maximum power coefficient; η_{trans} is the transmission system efficiency ($\eta_{trans} \in (0,1)$); η_{gen} is the electrical conversion efficiency of the generator ($\eta_{gen} \in (0,1)$).

The total output power of the wind-solar hybrid charging system is the sum of the actual output power of the PV system and the actual output power of the wind turbine. Considering the complementary nature of wind and solar resources, the total system output power can be expressed as:

$$P_{hybrid,act} = P_{PV,act} + P_{WT,act} \quad \#(9)$$

The selected Tulip wind turbine for this study can operate effectively even at temperatures as low as -40°C and under significant wind speed variations. It can quickly find the optimal pitch angle and tip-speed ratio to ensure maximum power output, thereby guaranteeing high wind energy utilization. Consequently, the selected wind turbine meets the requirements for deployment in the Antarctic region.

2.3 Estimation of Supercapacitor and Battery Output

Hybrid output systems combining supercapacitors and lithium batteries are predominantly utilized in applications requiring high-power delivery, such as new energy vehicles. However, their application in extremely low-temperature environments like Antarctica remains limited. Consequently, existing hybrid power supply architectures are unable to resolve the challenge of providing long-term, reliable power for seismic acquisition stations operating in remote polar field conditions.

Table III Research Status of Output Characteristics for Existing Supercapacitor-Lithium Battery Hybrid Systems

	Description	Applicable Scenarios	Operating Temperature	Output Power Range
[12]	Only when the voltage of the supercapacitor drops below that of the battery will the battery start to supply power in coordination.	Formula Hybrid Racing Car	Room Temperature (25°C)	6.5KW
[13]	lithium iron phosphate batteries and supercapacitors, designs hybrid energy storage topology and control strategies	Marine Electric Propulsion Systems	Room Temperature (25°C)	20MW
Proposed	Utilizing the low-temperature resistance of supercapacitors, extend the power supply time of the instrument	Antarctic Power Supply (Low-Energy Sensors)	Low Temperature (-40°C~20°C)	200w

Estimation of Supercapacitor and Battery Output

3.1. Battery Model

The SOC directly determines the energy management efficiency and operational reliability of the power supply system. This paper selects a second-order RC model for SOC estimation[14], which can more accurately capture the dynamic polarization characteristics of the battery compared to a first-order RC model[15].

$$U(t) = E_{OCV}(SOC, T) - I(t)R_0 - U_{C1}(t) - U_{C2}(t) \quad (10)$$

Where:

$U(t)$: Terminal voltage at time t (Unit: V)

$E_{OCV}(SOC,T)$: Open-circuit voltage under the coupling of SOC and temperature T (Unit: V)
 $I(t)$: Charge/discharge current at time t (Unit: A; positive for discharge, negative for charge)
 $U_{C1}(t)$ and $U_{C2}(t)$: Voltages across the fast/slow polarization capacitors (Unit: V), representing the voltage drop due to polarization effects.
Incorporating a correction for the low-temperature Antarctic environment:

$$E_{OCV}(SOC,T) = E_0 + K_1 \cdot SOC + K_2 \cdot \exp(-\lambda \cdot SOC) - \alpha(T + (60)) \quad (11)$$

The polarization resistances R1 and R2 decrease as SOC increases and increase as temperature decreases. The polarization capacitances C1 and C2 are less affected by temperature, maintaining relatively stable values.

Ohmic Internal Resistance R0 considering low temperature:

$$R_0(T) = R_0(25^\circ\text{C}) \times [1 + \beta(T + 60)] \quad (12)$$

($\beta=0.015/^\circ\text{C}$, Time constants: $\tau_1 = 30\text{-}300\text{ s}$, $\tau_2 = 1\text{-}10\text{ s}$)

3.2. Supercapacitor Model

C: Nominal capacitance Considering the linear voltage characteristic:

$$C = \frac{\Delta Q}{V_{max} - V_{min}} \quad (13)$$

R_{esr} : Equivalent series resistance

$$V_{sc} = \frac{1}{C} \int i_{sc} dt + R_{esr} i_{sc} \quad (14)$$

The State of Health (SOH) of the supercapacitor is defined as the ratio of its actual capacity to its nominal capacity, reflecting its degree of aging degradation[16], [17]. Combining equation (14) with the capacity calculation formula, the SOH evaluation model is established:

$$SOH = \frac{C_{act}}{C_{nom}} = \frac{\frac{\Delta Q_{act}}{V_{max} - V_{min}}}{C_{nom}} \quad (15)$$

Where:

C_{act} is the actual capacitance of the supercapacitor (Unit: F); C_{nom} is the nominal capacitance (Unit: F); ΔQ_{act} is the total charge that can be stored/delivered under actual operating conditions (Unit: C), calculable by current integration:

$$Q_{act} = \int_0^t I_C(t) dt \quad (16)$$

This evaluation model directly reflects the health state of the supercapacitor through its actual capacity degradation, providing a theoretical basis for the maintenance scheduling and lifetime prediction of the energy storage system.

3.3. PID-based Thermal Control

To maintain lithium batteries at a constant operating temperature in polar environments, this study proposes a collaborative temperature control strategy based on distributed silicon rubber heating patches and an adaptive incremental digital PID algorithm[18], [19], [20]. The system utilizes an array of high-precision PT1000 temperature sensors embedded within the battery

modules to monitor thermal status in real-time. Heating power decisions are generated according to a dynamic thermodynamic model, with the core design outlined as follows:

$$\Delta u(k) = K_p[e(k) - e(k-1)] + K_i T_s e(k) + K_d \frac{e(k) - 2e(k-1) + e(k-2)}{T_s} \quad \#(17)$$

IV. Overall Design of the System

The structural block diagram of the wind-solar hybrid power supply unit is shown in the figure below. The system comprises a wind turbine, photovoltaic panels, a power composite output controller, batteries, heating and temperature control modules, and can be broadly divided into three main parts: the power management module, the control module, and the power supply unit. The specific designs are as follows:

The power management module incorporates wind-solar hybrid power generation. It converts the electrical output through DC/DC conversion and then distributes it either to charge the batteries or to power the heating and temperature control module, which maintains a constant temperature within the equipment control box. This module ensures that the wind turbine and photovoltaic panels can complementarily convert wind and solar energy into electrical power to supply the entire unit. Considering the portability required for scientific research operations in polar regions, this project selects small-scale wind turbines and compact photovoltaic panels, as shown in Figure 2.

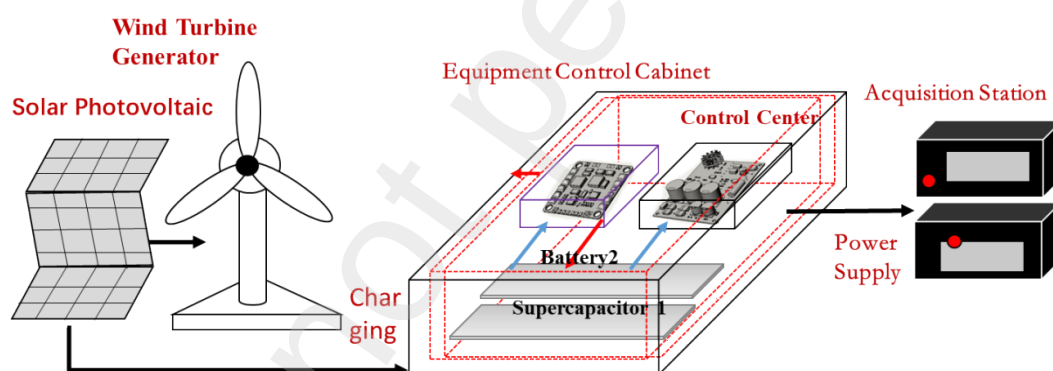


Figure 2 Schematic Diagram of the Power Supply System

4.1 Hardware Design

The control system is built around the STM32F407VET6 microcontroller as the core, forming a minimum system that serves as the main control unit for the entire wind-solar hybrid power supply system[21], [22]. It is responsible for the integrated management of energy conversion control, power distribution, data acquisition and storage, and protection functions. The core control logic is as follows: through preset control algorithms, it performs DC/DC rectification and voltage regulation on the unstable AC power output from the wind turbine and the DC power output from the photovoltaic panels[23]. The processed, stable DC power is then intelligently distributed based on load requirements and battery status, supplying both the energy storage unit and the heating/temperature control module to ensure the proper operation of core equipment in the extreme polar cold. Simultaneously, the system integrates functional modules for voltage/current signal acquisition,

battery pack charge/discharge switching, monitoring of ambient and internal equipment temperatures, and auxiliary protection (over-current, over-voltage, over-temperature, and under-voltage). It utilizes high-precision sensors to acquire key operational parameters in real-time—such as input/output voltages, charge/discharge currents, energy storage unit voltage, ambient temperature, and internal control box temperature—and stores the raw data along with system status information synchronously to an SD card. This enables comprehensive, full-cycle data logging of the system's operation process in the extreme Antarctic environment, providing essential data support for subsequent performance analysis and optimization, as shown in Figure 3. The physical object diagram is shown in Figure 4.

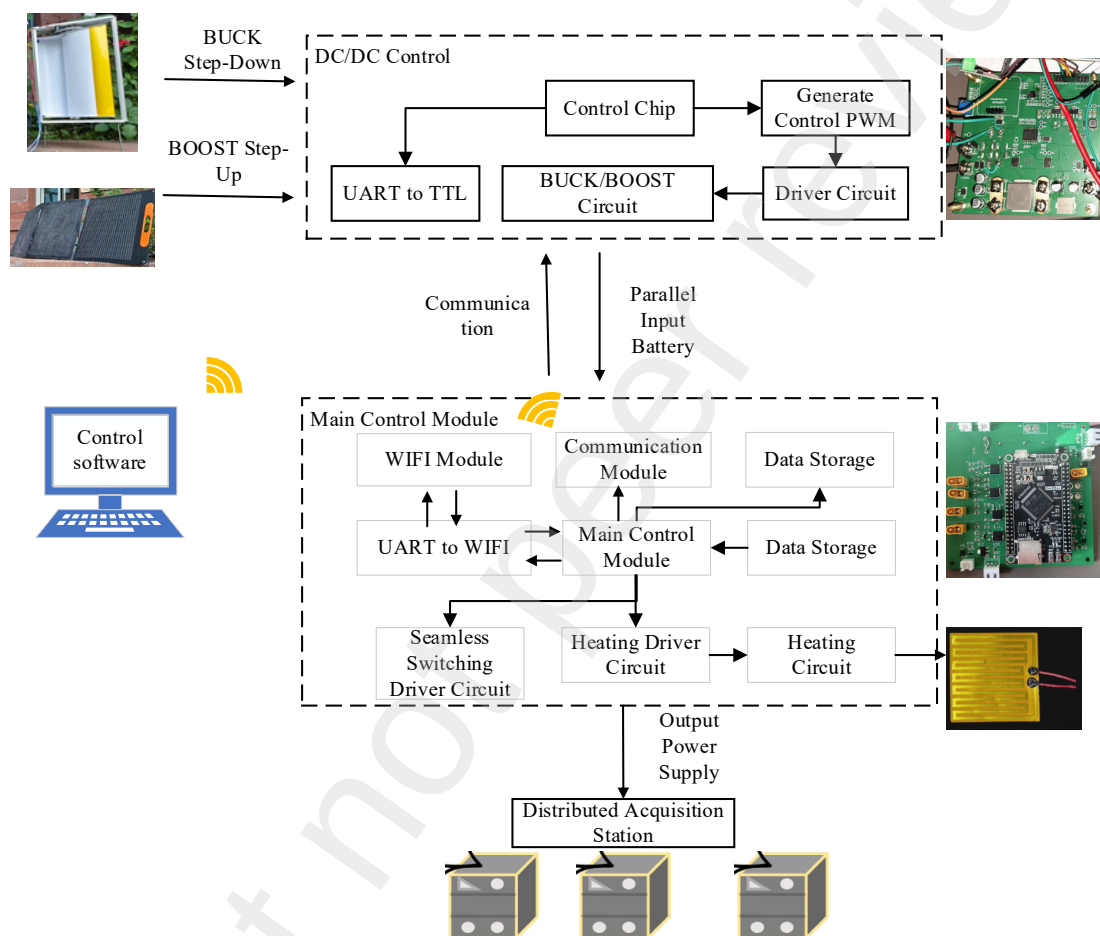


Figure 3 Overall Structure Block Diagram of Supply Device

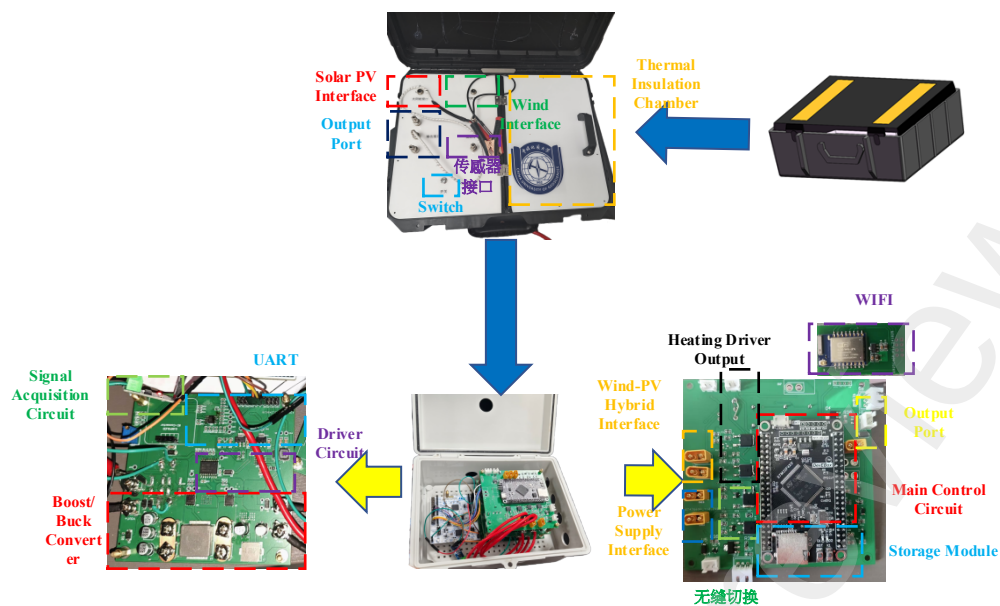


Figure 4 Physical Prototype Diagram

4.2 software Design

This fig outlines the software workflow for a wind-solar hybrid power supply system the process begins with system initialization (figure 5), followed by configuring the WiFi module to serial port communication; the system then continuously checks for a serial port interrupt—if triggered, it collects real-time operational data (current, voltage, temperature) and calculates the SOC and SOH of the energy storage unit before concluding this sub-process, while concurrently monitoring ambient and internal temperatures alongside current levels: if the detected value is less than 0 (indicating sub-zero temperatures or abnormal current), the supercapacitor activates to output power for heating, and for temperatures ≥ 0 , a hybrid heating mode (using PID control for constant-temperature heating) is initiated, paired with seamless switching output from the supercapacitor-lithium battery bank to maintain stable power supply[20]; finally, the system adjusts the wind-solar DC/DC converters to operate in constant-voltage and constant-current input modes for efficient energy harvesting, and the workflow loops back to re-detect environmental parameters for continuous operation.

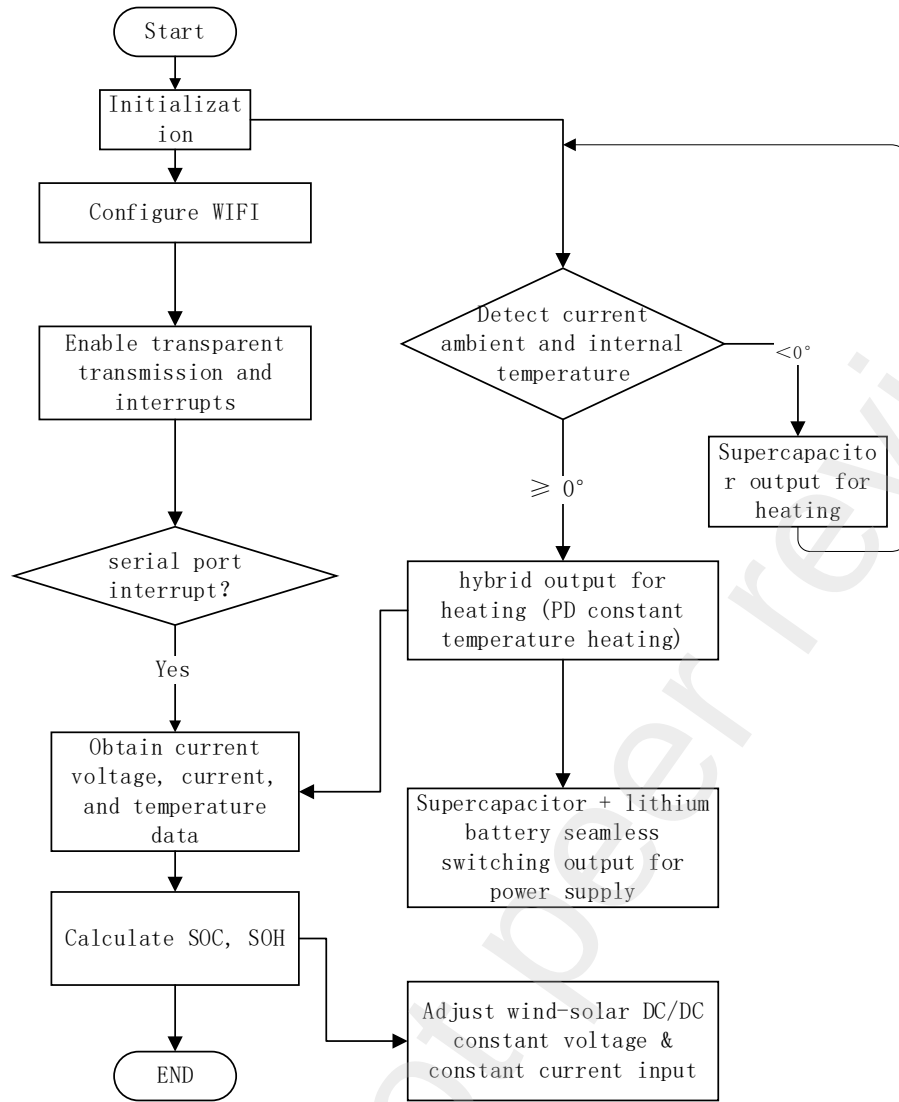


Figure 5 Software Flow Chart

V.Key Technologies

5.1. Control Method for Parallel DC/DC Input from Solar and Wind Sources

A. Control Method for Parallel DC/DC Output

Figure 6 shows the equivalent circuit diagram of the integrated wind-solar power supply based on the parallel output control of DC/DC converters. These two DC/DC converters are the DC control power stages following the photovoltaic panels and the wind turbine, respectively. As illustrated in Figure 6, for ease of analysis, let r represent the sum of internal resistances from switching devices, diodes, inductors, etc. To ensure the normal power output of both DC/DC converters, this paper adopts a control strategy where the converters operate in parallel: one converter is set to Constant Voltage (CV) control mode, while the other operates in Constant Current (CC) control mode. This configuration allows the converter in CC mode to act as the load for the converter in CV mode. Based on the equivalent circuit diagram, the average state-space model from input to output can be derived by solving the differential equations, as represented by the block diagram in Figure 7. On this basis, the PI control parameters for the voltage and current loops of each converter are designed.

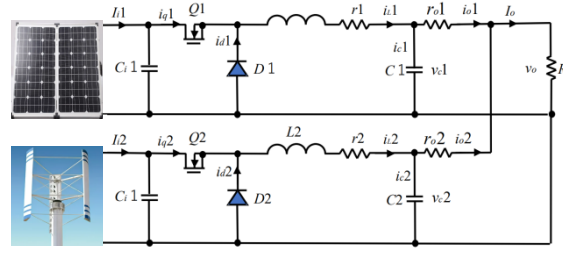


Figure 6 Equivalent Circuit Diagram of Combined Wind-PV Power Supply Based on Parallel Output Control of DC/DC Converters

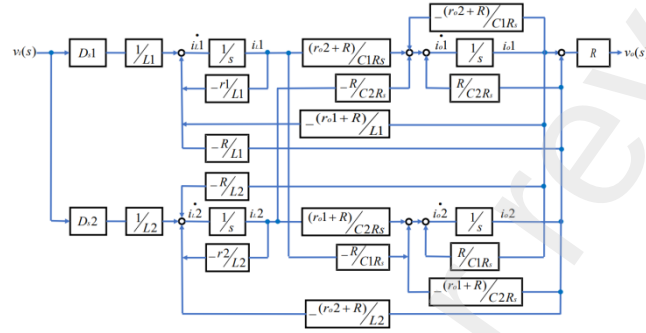


Figure 7 Block Diagram of Average State Equations for Parallel Circuit of DC/DC Converters

Figure 8 presents the state machine diagram corresponding to the average state equations of the parallel DC/DC converters, delineating the system's operational state transitions while interfacing with the earlier-established CV/CC control architecture and average state-space model. The framework encompasses five distinct states: 1.StateMInit(), StateMWait() , StateMRise() , StateMRun() , and StateMErr() .

The system commences at StateMInit()—where initial conditions for the state-space model are initialized—before transitioning to StateMWait(). Once the preset waiting duration expires, it advances to StateMRise(). Upon successful completion of the start-up sequence, the system enters StateMRun(), where the CV/CC control modes regulate steady-state operation via the state-space model's differential equation solutions. The occurrence of a fault triggers a shift to StateMErr(), while fault resolution resets the system to StateMWait() for reinitialization of control parameters.

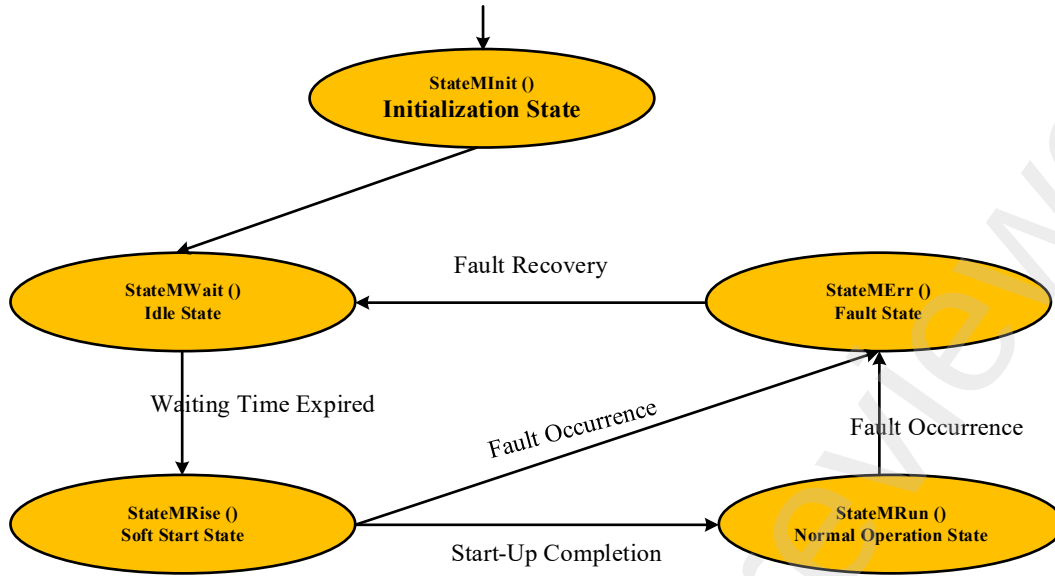


Figure 8 Block Diagram of Average State Equations for Parallel Circuit of DC/DC Converters

Due to the unique climatic conditions of Antarctica, both wind and solar energy exhibit significant fluctuations. Wind power output, in particular, varies considerably and is difficult to regulate predictably. Therefore, this study utilizes the relatively stable solar power as a CV source and wind power as a CC source, with their outputs combined in parallel.

The control objective for the solar DC/DC converter is to maintain a stable output voltage. Its PI control equation is as follows:

$$d_{pv}(t) = K_{p,v} \cdot e_v(t) + K_{i,v} \cdot \int_0^t e_v(\tau) d\tau \quad (18)$$

$$e_v(t) = V_{ref} - V_{out}(t) \quad (19)$$

Where: $e_v(t)$ is the voltage error;

$d_{pv}(t)$ is the duty cycle of the solar converter;

$K_{p,v}$ and $K_{i,v}$ are the proportional and integral coefficients of the voltage control loop, respectively.

The wind energy DC/DC converter aims to output a constant current. Its PI control equation is as follows:

$$d_{wind}(t) = K_{p,i} \cdot e_i(t) + K_{i,i} \cdot \int_0^t e_i(\tau) d\tau \quad (20)$$

$$e_i(t) = I_{ref} - I_{L2}(t) \quad (21)$$

Where:

$e_i(t)$ is the current error; $d_{wind}(t)$ is the duty cycle of the wind energy converter; $K_{p,i}$ and $K_{i,i}$ are the proportional and integral coefficients of the current control loop, respectively.

Figure 9 presents the corresponding simulation results of the hybrid power supply. The obtained waveform simulation results demonstrate that the designed PI control parameters are appropriate. Although the output current of one side changes rapidly, the overall power supply system maintains stable output voltage and current. The total output current equals the sum of the output currents from the two converters, achieving the intended power superposition.

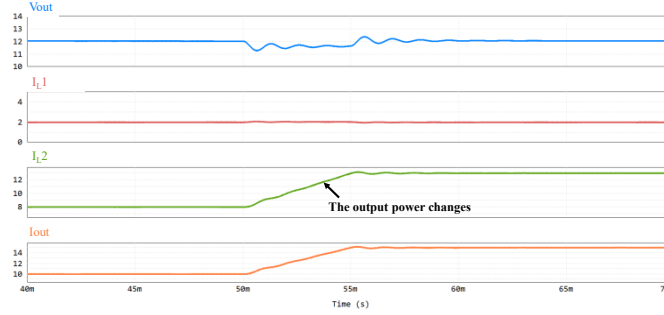


Figure 9 Simulation Results of Wind-PV Power Supply Based on Parallel Output Control of DC/DC Converters

The total output current and power of the parallel system are as follows:

$$I_{out}(t) = I_{L1}(t) + I_{L2}(t) \quad (21)$$

$$P_{out}(t) = V_{out}(t) \times I_{out}(t) \quad (22)$$

Since the solar source maintains a constant voltage, the total power increases linearly with the wind source current I_{L2} .

The system utilizes a Buck converter, whose inductor current dynamics are governed by:

$$L \frac{dI_L}{dt} = d \cdot V_{in} - V_{out} \quad (23)$$

Combined with PI control, the closed-loop transfer function can be expressed as:

$$G_{closed}(s) = \frac{K_p s + K_i}{L s^2 + (R + K_p) s + K_i} \quad (24)$$

Here, R represents the equivalent load resistance. Pole placement techniques are necessary to ensure system stability.

1. The solar-powered constant voltage source rapidly adjusts its duty cycle via the voltage loop PI controller, effectively suppressing the impact of load fluctuations on the output voltage.

2. The wind-powered constant current source tracks the reference current through its current loop PI controller. This strategy allows for power fluctuations while limiting current surges.

3. During parallel operation, output impedance matching is ensured between the two converters (low impedance for the solar source, high impedance for the wind source) to prevent circulating currents.

Figure 10 presents the Bode diagram of the parallel DC/DC converter system, characterizing the frequency-domain performance of the voltage loop, current loop, and overall closed-loop system (consistent with the PI-controlled CV/CC strategy above). The magnitude plot shows that the voltage loop maintains high gain at low frequencies—supporting its role in suppressing load-induced voltage fluctuations—while the current loop exhibits moderate gain, aligning with its current-tracking function. The closed-loop system combines these responses, ensuring stability. The phase plot further illustrates that the voltage loop introduces steeper phase shift at higher frequencies,

while the closed-loop system retains sufficient phase margin to prevent oscillations—critical for avoiding circulating currents and ensuring robust parallel operation.

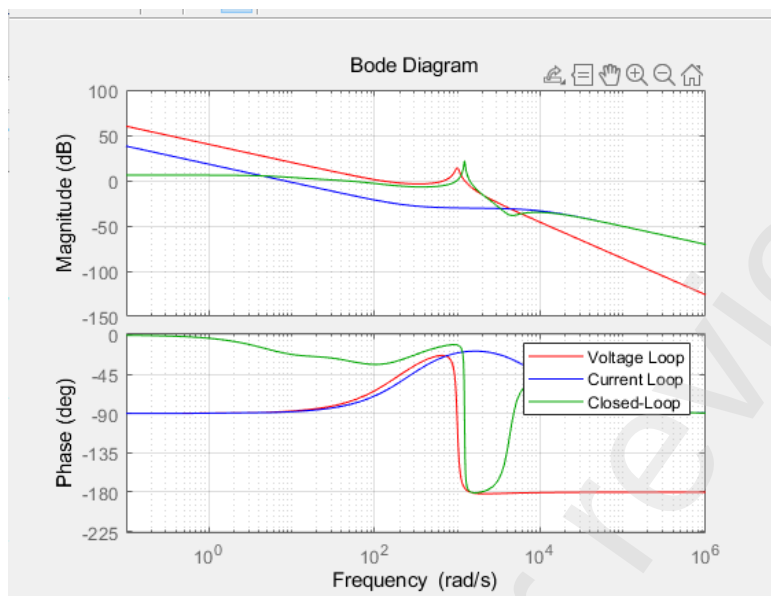
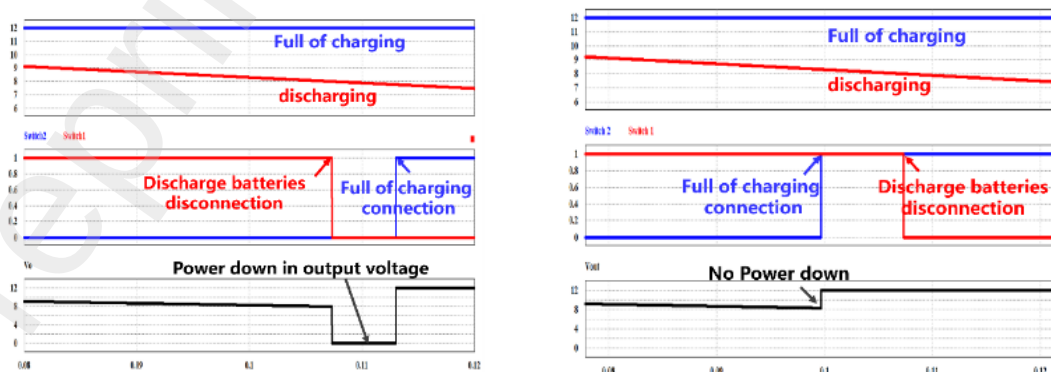


Figure 10 Bode Plot of the Parallel System

5.2. Output Switching

Conventional battery switching methods involve disconnecting the active power supply battery from the load before connecting the backup battery. This approach creates a period without power supply during the switching interval, leading to a rapid decrease in output voltage and compromising power supply stability. To address the issue of unstable module output voltage, a dual MOSFET parallel switching circuit is designed. Leveraging the characteristic that during parallel battery operation only the side with the higher voltage supplies power, this paper proposes a seamless switching control strategy that prevents rapid voltage drops.

Figure 11 shows the simulation results based on the proposed battery seamless switching control process. As depicted in Figure 11(a), even with a very short controlled power-off duration, the output voltage still experiences a brief interruption, adversely affecting the normal operation of the load. Figure 11(b) presents the simulation results of the method proposed in this project, which involves first connecting the backup battery before disconnecting the discharging battery.

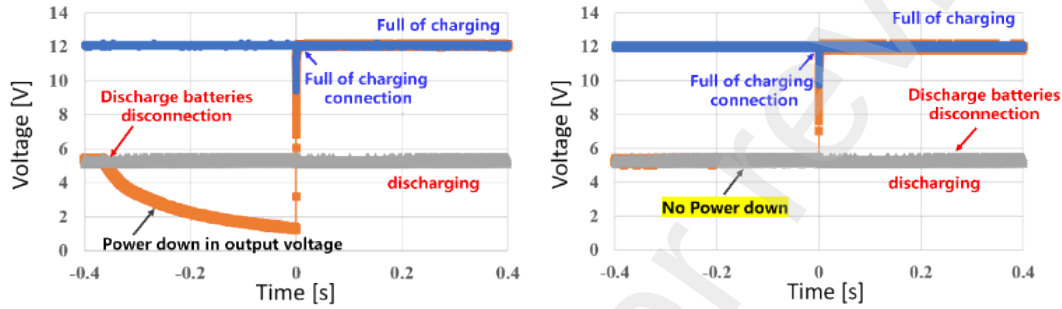


(a) Conventional Control Method

(b) Proposed Control Method

Fig.11 Simulation Results of Seamless Battery Switching Control Method

The experimental results are shown in Figure 12. Figure 12(a) displays the outcome using the conventional battery switching method of "break-before-make." During the battery switching process, the output voltage experiences a brief power interruption. In contrast, Figure 12(b) shows the result of employing the innovative "make-before-break" battery switching control strategy. Apart from a minor voltage transient at the moment of battery connection, the output voltage remains stable. The results demonstrate that the method proposed in this scheme effectively avoids output voltage transients and sags, thereby providing a stable power supply to the load.



(a) Conventional Control Method

(b) Proposed Control Method

Fig.12 Test Results of Seamless battery switching Control Method

5.3.Cold-Start Strategy with Temperature Management Mechanism

To address the challenge of initiating energy storage systems in extremely cold environments, this paper proposes a hybrid parallel topology combining supercapacitors and lithium batteries. A three-phase cold-start protocol is designed for operation at -40°C :

1. Supercapacitor Transient Response Phase ($t < 30$ s): The supercapacitor supplies power.
2. Lithium Battery Pre-activation Phase: Residual energy from the supercapacitor is used to trigger PTC phase-change heating.
3. Hybrid Power Supply Phase: Voltage fluctuation is controlled within $\pm 1\%$.

Discharge tests were conducted on a lithium-ion battery, a supercapacitor, and a hybrid energy storage module at -30°C . Using a constant discharge current of 1A, the time taken for the voltage of each unit to drop from 12V to 8V was measured. The discharge curves are shown in the accompanying figure. The results indicate that under low-temperature conditions, both the supercapacitor and the hybrid module can discharge normally, whereas the lithium-ion battery fails to operate correctly, remaining in a low-voltage state. However, due to the supercapacitor's limited nominal capacitance of 160 F, its discharge duration was only 620 seconds. In contrast, the hybrid module's effective capacity increased to 1700 F, extending the discharge duration to 4500 seconds. This demonstrates that the hybrid module not only possesses low-temperature tolerance but also ensures adequate energy density, showing promising potential for application in Antarctic environments.

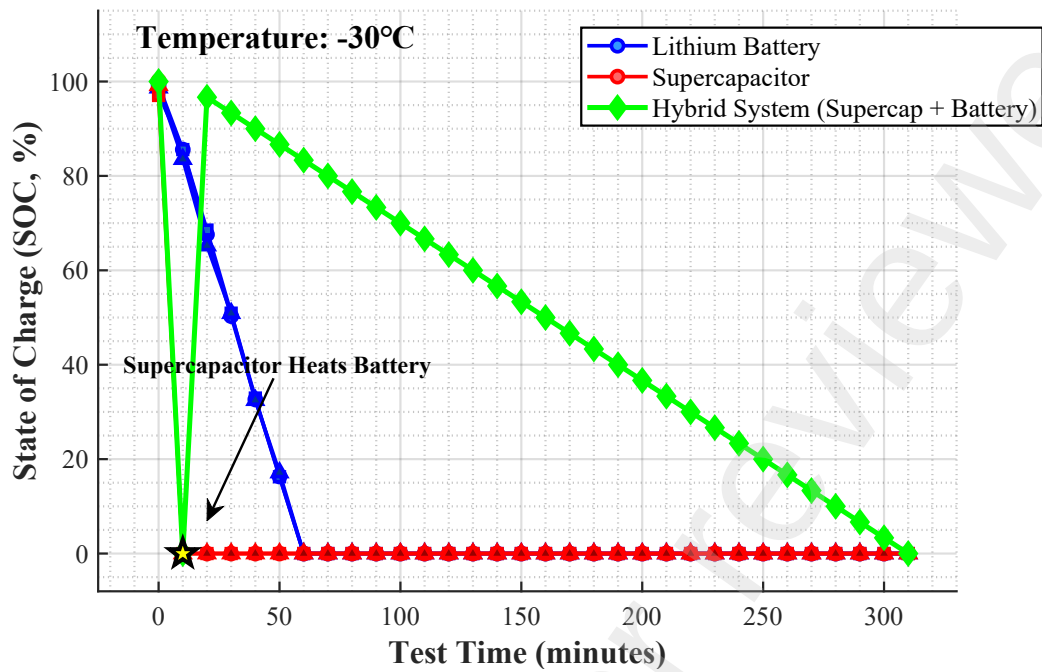


Figure 13 Experimental Diagram of Endurance Time Before and After Cold Start

VI. Experiments

Tests were conducted in Antarctica from January 11 to February 17, 2025, with a cumulative testing duration of 127 hours. The testing scope included:

1. Long-term operational stability tests of the input and output for the wind-solar hybrid power supply unit.
2. Parallel charging tests measuring voltage, current, and their combined output for the wind-solar hybrid system.
3. Thermal insulation performance tests of the equipment control box under low-temperature conditions.



Fig. 14. On-Site Field Test Image in Antarctica

The results of these tests are shown in the corresponding figures below. The data verifies the overall system stability and sustained operational capability over extended periods.

Figure 15-Figure 17 presents four experimental test waveforms for the wind-solar hybrid power supply system, collected during a 127-hour field test in Antarctica (January 11–February 17, 2025)—a setting characterized by sub-zero temperatures (consistently below -20°C) and variable environmental conditions, which stress-test the system's performance under extreme operational constraints.

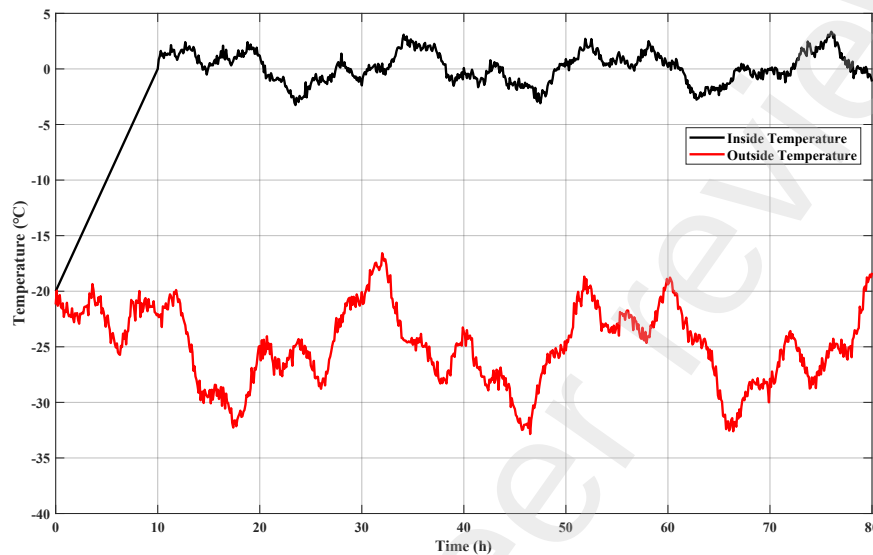


Figure 15. Temperature Test Diagram

Figure 15 underscores the equipment control box's robustness: the internal temperature stabilizes around 5°C despite ambient temperatures plummeting to $\sim -20^{\circ}\text{C}$ —a critical outcome, as Antarctic cold could otherwise impair circuit functionality. This result directly confirms the control box's ability to maintain operational temperatures, a core test objective.

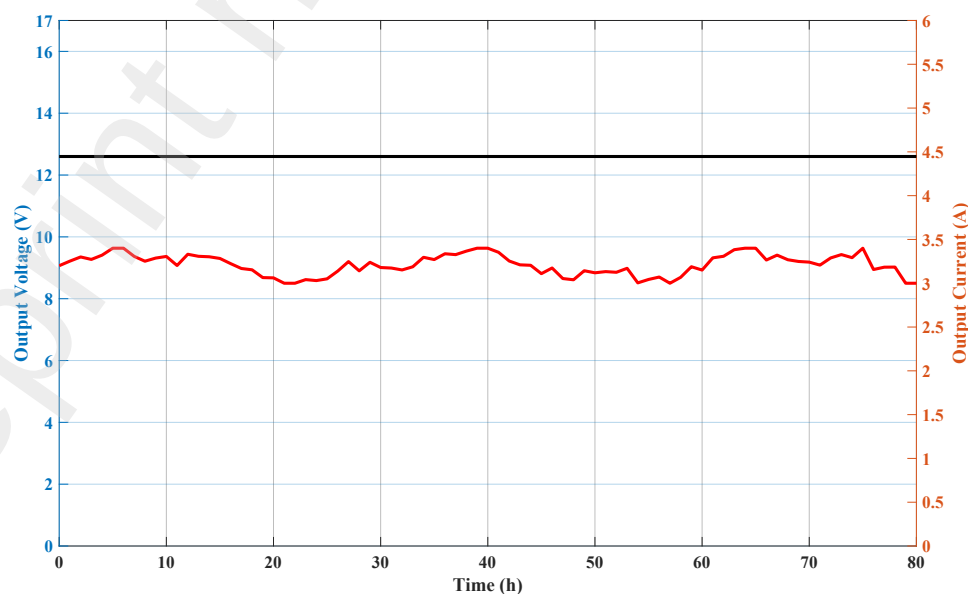


Figure 16. Wind-Solar Hybrid Parallel Constant Voltage Test Diagram

Figure 16 shows DC output voltage oscillating slightly around the target value: this stability, achieved despite Antarctica's variable renewable energy input (e.g., weak solar irradiance in polar winter), corroborates the parallel system's voltage regulation performance under non-ideal source conditions.

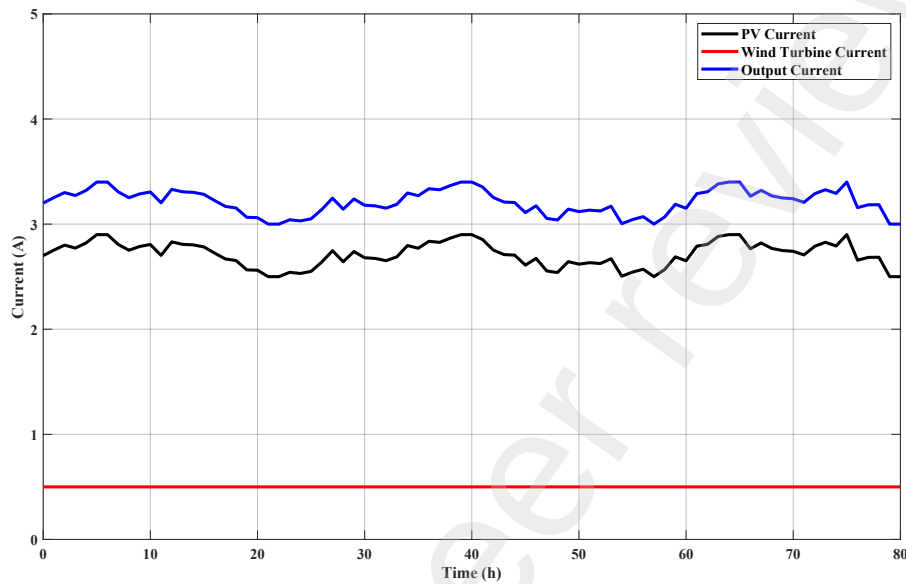


Figure 17. Wind-Solar Hybrid Parallel Constant Current Test Diagram

Figure 17 displays solar current, wind current, and combined output current: the steady wind current reflects the CC controller's ability to track references even in cold-induced component parameter shifts, while coordinated current fluctuations verify the impedance-matching framework's effectiveness at preventing circulating currents—an essential safeguard for system longevity in remote Antarctic deployments. These results confirm the system's suitability for extreme low-temperature, off-grid environments.

VII. Conclusion

This paper proposes an efficient and stable wind-solar-storage coupled power supply technology based on supercapacitors and lithium batteries for low-temperature environments. The main contributions of this technology are as follows:

The limitations of power supply in Antarctica were analyzed, leading to the proposal of a multi-energy coupling strategy.

1. A multi-source parallel input system was established at the input end. Through a coordinated operation strategy, solar panels and wind turbines achieve complementary power output. Compared

to previous strategies, the energy utilization rate is increased by 60%.

2. A hybrid output system combining supercapacitors and lithium batteries was constructed at the output end. To address the performance degradation of batteries at low temperatures, supercapacitors are utilized for preheating.

3. Practical testing in Antarctica demonstrated that the power supply duration for the overall instrument system was extended by 7.25 times. The proposed power supply technology reduces voltage fluctuations, effectively mitigates power instability caused by renewable energy intermittency and extreme cold, provides crucial support for the design of efficient and stable power systems in Antarctic environments, and ensures the reliable operation of geophysical instruments.

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